

TSUNAMI AND COASTAL HAZARD MANAGEMENT

TSUNAMI GENERATION CHAIN → COASTAL-HAZARD FIREWALL → SOURCE-TO-IMPACT MATRIX → DETECTION-CONFIRMATION RAIL → ITEWC AUTHORITY MAP → WARNING-UNCERTAINTY LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

TSUNAMI GENERATION CHAIN

TSUNAMI GENERATION CHAIN

SUDDEN WATER DISPLACEMENT

VOLCANIC ACTIVITY

LONG WAVES

COASTAL TRANSFORMATION

RUN-UP + INUNDATION + RETURN FLOW

RISK CONDITION / INPUT

- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- LONG WAVES → COASTAL TRANSFORMATION
- RUN-UP + INUNDATION + RETURN FLOW

LAST-MILE / INCLUSION / OUTCOME LIMIT

- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONG WAVES → COASTAL TRANSFORMATION; RUN-UP + INUNDATION + RETURN FLOW.

01

STAGE 01

COASTAL-HAZARD FIREWALL

COASTAL-HAZARD FIREWALL

SUDDEN DISPLACEMENT

GRAVITATIONAL CYCLE

STORM SURGE

CYCLONE WIND + PRESSURE

DIFFERENT MONITORING AND RESPONSE LOGIC

HAZARD / ACTOR / INSTRUMENT

TSUNAMI → sudden displacement
TIDE → gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

STORM SURGE → cyclone wind + pressure
RULE → different monitoring and response logic

WARNING / IMPLEMENTATION / OUTCOME LIMIT

TSUNAMI → sudden displacement
TIDE → gravitational cycle

HAZARD / ACTOR / INSTRUMENT

- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

- STORM SURGE → cyclone wind + pressure
- RULE → different monitoring and response logic

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE → cyclone wind + pressure; RULE → different monitoring and response logic.

02

STAGE 02

SOURCE-TO-IMPACT MATRIX

SOURCE-TO-IMPACT MATRIX

MECHANISM DEPTH DISPLACEMENT

DISTANCE AND OCEAN CONFIGURATION

BATHYMETRY TOPOGRAPHY LAND USE

EXPOSURE VULNERABILITY AND ACTION

HAZARD / ACTOR / INSTRUMENT

SOURCE → mechanism depth displacement
PATH → distance and ocean configuration

DECISIVE RISK OR MANDATE DISTINCTION

COAST → bathymetry topography land use
OUTCOME → exposure vulnerability and action

WARNING / IMPLEMENTATION / OUTCOME LIMIT

SOURCE → mechanism depth displacement
PATH → distance and ocean configuration

HAZARD / ACTOR / INSTRUMENT

- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

DECISIVE RISK OR MANDATE DISTINCTION

- COAST → bathymetry topography land use
- OUTCOME → exposure vulnerability and action

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COAST → bathymetry topography land use; OUTCOME → exposure vulnerability and action.

03

STAGE 03

DETECTION-CONFIRMATION RAIL

DETECTION-CONFIRMATION RAIL

1 SEISMIC EVENT ANALYSIS

2 PRE-RUN MODEL SCENARIO

3 SEA-LEVEL MONITORING

4 UPDATED BULLETIN

THREAT STATUS

HAZARD / RISK / RESILIENCE CONCEPT

- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

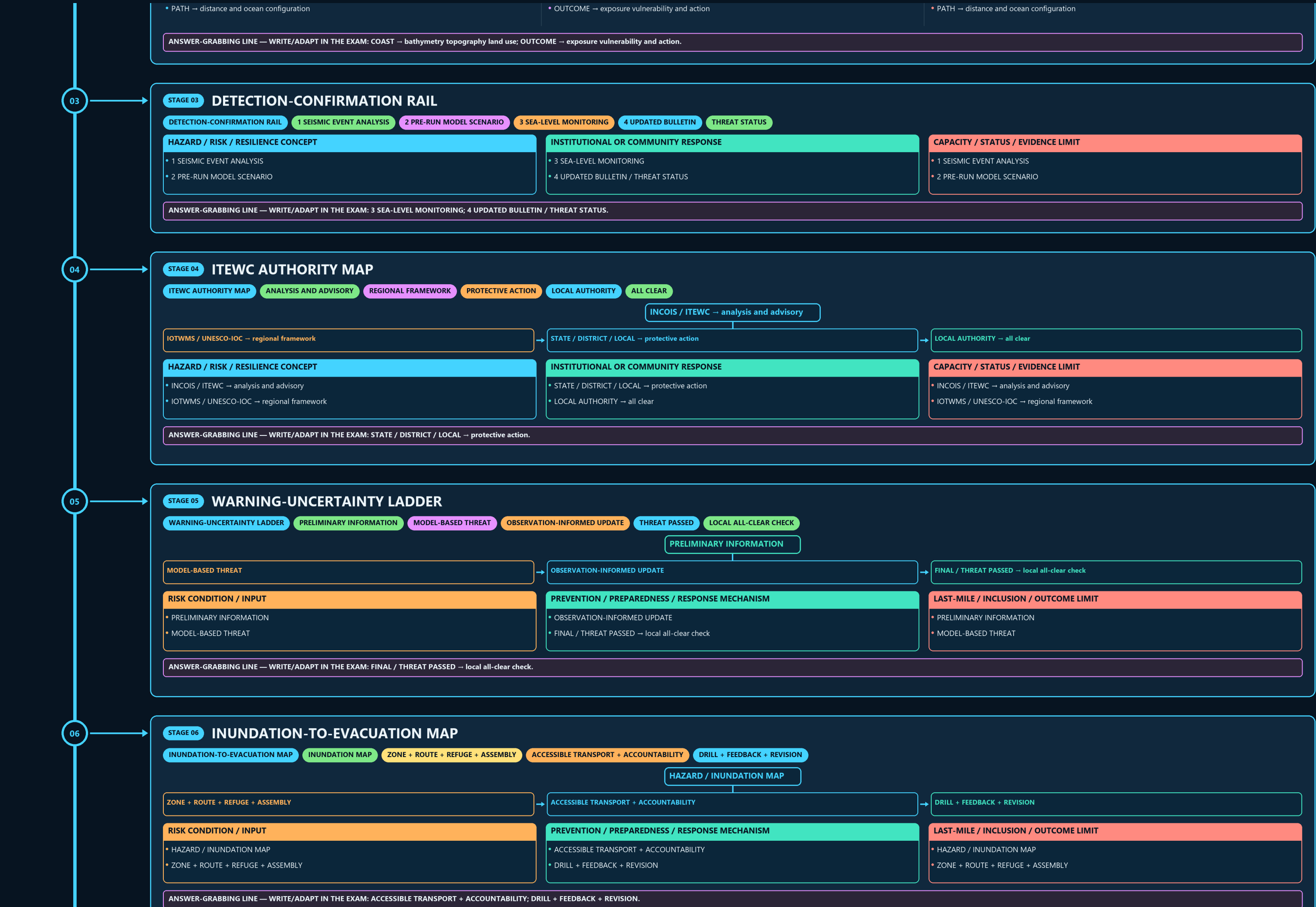
INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 SEA-LEVEL MONITORING
- 4 UPDATED BULLETIN / THREAT STATUS

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 SEA-LEVEL MONITORING; 4 UPDATED BULLETIN / THREAT STATUS.



05

STAGE 05

WARNING-UNCERTAINTY LADDER

WARNING-UNCERTAINTY LADDERPRELIMINARY INFORMATIONMODEL-BASED THREATOBSERVATION-INFORMED UPDATETHREAT PASSEDLABEL ALL-CLEAR CHECK

PRELIMINARY INFORMATION

MODEL-BASED THREAT

OBSERVATION-INFORMED UPDATE

FINAL / THREAT PASSED → local all-clear check

RISK CONDITION / INPUT

• PRELIMINARY INFORMATION

• MODEL-BASED THREAT

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

• OBSERVATION-INFORMED UPDATE

• FINAL / THREAT PASSED → local all-clear check

LAST-MILE / INCLUSION / OUTCOME LIMIT

• PRELIMINARY INFORMATION

• MODEL-BASED THREAT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FINAL / THREAT PASSED → local all-clear check.

06

STAGE 06

INUNDATION-TO-EVACUATION MAP

INUNDATION-TO-EVACUATION MAPINUNDATION MAPZONE + ROUTE + REFUGE + ASSEMBLYACCESSIBLE TRANSPORT + ACCOUNTABILITYDRILL + FEEDBACK + REVISION

HAZARD / INUNDATION MAP

ZONE + ROUTE + REFUGE + ASSEMBLY

ACCESSIBLE TRANSPORT + ACCOUNTABILITY

DRILL + FEEDBACK + REVISION

RISK CONDITION / INPUT

• HAZARD / INUNDATION MAP

• ZONE + ROUTE + REFUGE + ASSEMBLY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

• ACCESSIBLE TRANSPORT + ACCOUNTABILITY

• DRILL + FEEDBACK + REVISION

LAST-MILE / INCLUSION / OUTCOME LIMIT

• HAZARD / INUNDATION MAP

• ZONE + ROUTE + REFUGE + ASSEMBLY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESSIBLE TRANSPORT + ACCOUNTABILITY; DRILL + FEEDBACK + REVISION.

06

STAGE 06 INUNDATION-TO-EVACUATION MAP

INUNDATION-TO-EVACUATION MAP INUNDATION MAP ZONE + ROUTE + REFUGE + ASSEMBLY ACCESSIBLE TRANSPORT + ACCOUNTABILITY DRILL + FEEDBACK + REVISION

HAZARD / INUNDATION MAP

ZONE + ROUTE + REFUGE + ASSEMBLY

ACCESSIBLE TRANSPORT + ACCOUNTABILITY

DRILL + FEEDBACK + REVISION

RISK CONDITION / INPUT

- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- ACCESSIBLE TRANSPORT + ACCOUNTABILITY
- DRILL + FEEDBACK + REVISION

LAST-MILE / INCLUSION / OUTCOME LIMIT

- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESSIBLE TRANSPORT + ACCOUNTABILITY; DRILL + FEEDBACK + REVISION.

07

STAGE 07 COASTAL PROTECTION PORTFOLIO

COASTAL PROTECTION PORTFOLIO SHELTERS WALLS BREAKWATERS WHERE JUSTIFIED DUNES MANGROVES FORESTS REEFS CRZ + RISK-SENSITIVE SITING LAYERED AND LOCATION-SPECIFIC

HAZARD / ACTOR / INSTRUMENT

ENGINEERED → shelters walls breakwaters where justified
NATURAL → dunes mangroves forests reefs

DECISIVE RISK OR MANDATE DISTINCTION

REGULATORY → CRZ + risk-sensitive siting
RULE → layered and location-specific

WARNING / IMPLEMENTATION / OUTCOME LIMIT

ENGINEERED → shelters walls breakwaters where justified
NATURAL → dunes mangroves forests reefs

HAZARD / ACTOR / INSTRUMENT

- ENGINEERED → shelters walls breakwaters where justified
- NATURAL → dunes mangroves forests reefs

DECISIVE RISK OR MANDATE DISTINCTION

- REGULATORY → CRZ + risk-sensitive siting
- RULE → layered and location-specific

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- ENGINEERED → shelters walls breakwaters where justified
- NATURAL → dunes mangroves forests reefs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY → CRZ + risk-sensitive siting; RULE → layered and location-specific.

08

STAGE 08 COASTAL ECONOMY CONTINUITY

COASTAL ECONOMY CONTINUITY SHUTDOWN ACCESS REOPENING BOATS LANDING CENTRES MARKETS VISITOR WARNING SIGNAGE ROUTE CONTINUITY AND SAFE RECOVERY

PORT / HARBOUR → shutdown access reopening

FISHERIES → boats landing centres markets

TOURISM → visitor warning signage route

UTILITIES / HOSPITAL → continuity and safe recovery

HAZARD / RISK / RESILIENCE CONCEPT

- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

INSTITUTIONAL OR COMMUNITY RESPONSE

- TOURISM → visitor warning signage route
- UTILITIES / HOSPITAL → continuity and safe recovery

CAPACITY / STATUS / EVIDENCE LIMIT

- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UTILITIES / HOSPITAL → continuity and safe recovery.

09

STAGE 09 TSUNAMI READY CYCLE

TSUNAMI READY CYCLE MAP RISK EVACUATION MAP + SIGNAGE EVALUATE INDICATORS RENEW READINESS COMMUNITY RECOGNITION NOT NATIONAL CERTIFICATION

MAP RISK → EDUCATE → WARN

EVACUATION MAP + SIGNAGE → DRILL

EVALUATE INDICATORS → RENEW READINESS

COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

RISK CONDITION / INPUT

- MAP RISK → EDUCATE → WARN
- EVACUATION MAP + SIGNAGE → DRILL

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- EVALUATE INDICATORS → RENEW READINESS
- COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MAP RISK → EDUCATE → WARN
- EVACUATION MAP + SIGNAGE → DRILL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVALUATE INDICATORS → RENEW READINESS; COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION.

